

Parameters optimization and performance evaluation of multiple tuned mass dampers to mitigate the vortex-induced vibration of a long-span bridge

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ABSTRACT

Vortex-induced vibration (VIV) is one of the primary dynamic problems of long-span bridges, which affects the safety and serviceability of bridges. The control and mitigation of VIV is a hotspot to researchers. Multiple tuned mass dampers (MTMD) consist of a set of tuned mass dampers (TMD) with varied frequencies and damping ratios, which is a common method for vibration control of structures. In this study, a concept of complex effective mass is proposed, which can entirely represent the dynamic effect of MTMD on bridges, and a derivative-based MTMD optimization process for VIV control of long-span bridges is presented. The VIV mitigation of a six-span continuous steel box-girder bridge is used to analyze the performance and mechanism of MTMD from the aspects of frequency domain and time domain. Optimal parameters of MTMD for suppressing VIV are obtained by the proposed algorithm. The results indicate that MTMD can significantly reduce the displacement amplitude of the beam due to VIV, and the performance of MTMD is better than that of single tuned mass damper (STMD) with the same mass ratio. The optimal performance of MTMD is independent of the mass distribution of the sub-TMDs. Moreover, the damping ratio of MTMD is much lower than that of STMD, but it can still help the vibration of the primary structure to decay faster. Complex mode theory and transfer function theory are used to explain this phenomenon. It indicates that the performance of MTMD does not depend on its damping ratio but is due to the phase interference between complex modes, and a one-way energy barrier between the bridge deck and MTMD can be formed, which prevents the transfer of vibration energy from MTMD to the primary structure after unloading. Finally, the stroke of MTMD in the control of long-span bridge VIV is analyzed, and a simplified formula is proposed.

1. Introduction

With the development of the economy and society, the requirements of engineering structure are no longer limited to the safety of the operation stage but more to ensure comfort. As an essential infrastructure for crossing rivers, straits, and canyons, long-span bridges play a vital role in the transportation network. In recent years, vortex-induced vibration (VIV) has been observed on many worldwide bridges, such as the Trans-Tokyo Bay Bridge in Japan [1], Volgograd Bridge in Russia [2], Xihoumen Bridge, Yingwuzhou Yangtze River Bridge and Humen Bridge in China [3]. Although VIV is a self-limited vibration, it generally occurs in a low wind speed range, which will affect driving safety, cause long-term fatigue damage and shorten the life of bridges [3–6].

There are two main methods of bridge VIV control, aerodynamic

measures and mechanical measures. For the former, it is mainly to eliminate the vortex shedding by optimizing the aerodynamic shape of bridge deck, installing attachments [7–9] (e.g., spoilers, guide vanes, failing plates) and suction-blowing devices [6]. However, due to the scale effect and approximate treatment in section test, it is not guaranteed that the aerodynamic measures can be fully effective on the real bridge. In contrast, mechanical measures seem simple and more reliable by installing mechanical devices to increase the damping ratio of the bridge. Viscous dampers, the most common method in the vibration control of buildings [10,11], are not so suitable for application in bridge engineering due to the reason that it is difficult to find two close positions with sufficient relative velocity on the bridge. Tuned mass damper (TMD), another main method of mechanical measures, can transfer the vibration energy of the bridge to the additional mass block by tuning and

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has become the preferred option for the beam vertical vibration control of long-span bridges.

TMD was first proposed as a spring-mass system by Frahm in 1909 [12]. Since the system has no energy dissipation device, it is only effective for the vibration of a single specific frequency. After that, Den Hartog [13,14] introduced the viscous damper into the system, which constituted the most common TMD configuration, and proposed a series of optimal TMD design formulae based on the fixed-point theory. However, the performance of STMD depends on frequency tuning and optimal damping, it will be significantly reduced when the primary structure and TMD change [15,16]. Many attempts have been made by researchers to weaken the mistuning problem of STMD, e.g., nonlinear TMD [17,18], adaptive-passive TMD [19–21], semi-active TMD [22–24]. Alternatively, a novel concept of multiple tuned mass dampers (MTMD) consisting of a set of sub-TMDs with varied frequencies was proposed by Igusa and Xu [25], and the fundamental characteristics of MTMD were further investigated by H. Yamaguchi [26] and M. Abé [27]. Subsequently, a large number of studies on MTMD have been conducted, mainly including optimization methods and performance evaluation. Park et al. [28] investigated the control effect and robustness of uniformly and linearly distributed MTMD for the vibration mitigation of a single-degree-of-freedom(SDOF) system subjected to harmonic and seismic excitation. Li et al. [29] proposed a gradient-based method for optimizing non-uniformly distributed MTMD and conducted a parametric analysis on the control effect and robustness. To enhance the robustness of MTMD, Kim et al. [30] proposed a nonlinear MTMD with coulomb dry friction and converted the nonlinear system into an equivalent linear system. It was demonstrated that the nonlinear MTMD has better robustness than the classical MTMD. The performance of MTMD in the vibration control of multi-degree-of-freedom(MDOF) structures has also attracted the interest of many researchers. Patil et al. [16] found that MTMD is quite effective and robust in the vibration mitigation of a benchmark building subjected to across wind loads. Daniel et al. [31] proposed to use MTMD to suppress the vibration of 3D irregular buildings and conducted a performance-based-design to minimize the total weight of MTMD based on gradient optimization. It was found that MTMD can mitigate both structural and nonstructural earthquake damage. Miguel et al. [32] proposed a hybrid stochastic/deterministic algorithm to determine the optimal parameters of MTMD with vertical and horizontal distribution for a benchmark building subjected to seismic excitation. To avoid exceeding the axial capacity of the upper floors, Nigdeli [33] and Elias [34] distributed the sub-TMDs along the heights of the building and compared the performance of the distributed MTMD with that of STMD. It was found that the distributed MTMD can also obtain effective passive control. More recently, MTMD has been used for the vibration control of the jacket-type offshore wind turbine subjected to multi-field coupling loads, such as wind, wave and current loads [35] and wind, wave and seismic loads [36]. The results of these studies confirm the superiority of MTMD in both control effect and robustness compared to STMD.

Since the VIV in bridges is always dominated by a single mode [37], STMD/MTMD is a good option for the mitigation of VIV. Huang et al. [38] investigated the performance of uniformly distributed MTMD for high-order VIV control of a long-span suspension bridge. Dai et al. [37,39] derived the equivalent damping ratio contributed by TMD based on empirical linear and nonlinear vortex force model and investigated the control effect and robustness of TMD for VIV mitigation. Xu et al. [40] and Dai et al. [41] compared the conventional TMD and inerter-based TMD for VIV control of long-span bridges, including the response reduction ratio of the bridge deck, the static stretching and stroke of TMD. So far, TMD has been applied in some real bridge engineering and proved to be effective in suppressing the VIV of long-span bridges(e.g., Trans-Tokyo Bay Bridge [1], Great East Belt Bridge [42], Chongqi Bridge [43], Hong Kong-Zhuhai-Macao Bridge [37], etc.). However, two secondary eigenmodes will be generated near the eigenmode when STMD is adopted, and the amplitude of which is

significantly higher than that of the eigenmode [14]. Thanks to the lock-in phenomenon of VIV, the VIV amplitude is dominated by the amplitude-frequency characteristic of a specific frequency band rather than a frequency point. Therefore, STMD is not a highly efficient scheme for VIV control since all of the mass is concentrated at one frequency point. In contrast, MTMD consists of a set of TMDs with varied frequencies, by optimizing and reasonably distributing the masses to a frequency range, and makes the vibration amplitude of the primary structure more uniform and lower within a specific frequency band than that of STMD. To the authors' best knowledge, the research on MTMD to mitigate the VIV of long-span bridges is rare.

This study aims to investigate the mechanism and performance of MTMD on the VIV control of long-span bridges. In Section 2, the governing equations of the bridge-MTMD system are derived, and the beam is simplified as a SDOF system in the modal space based on the orthogonality of the eigenvectors. A semi-empirical linear model is used to simulate the vortex-induced force acting on the bridge, and the transfer function of the generalized coordinate is obtained. A concept of complex effective mass is proposed, which can entirely represent the dynamic effect of MTMD on the bridge. In Section 3, a hybrid optimization process combining genetic algorithm and gradient algorithm is proposed to determine the optimal parameters of MTMD for the VIV control. In Section 4, a six-span continuous steel box-girder bridge is taken as a numerical example. The control effect of MTMD and the phase interference between complex modes of the bridge-MTMD system are studied in both frequency and time domain, and the energy dissipation path of the oscillation system is also analyzed. Moreover, the stroke of MTMD and its sensitivity to error in structural natural frequency are discussed. Finally, the conclusions of the present study are drawn in Section 5.

2. Dynamics of the bridge deck with MTMD

2.1. Governing equations of bridge-MTMD system

MTMD consists of a set of sub-TMDs with different parameters, and each sub-TMD is placed at different positions on the bridge deck, as shown in Fig. 1. The motion equations of the system can be expressed as

$$\bar{m}(x) \frac{\partial^2 y}{\partial t^2} + c(x) \frac{\partial y}{\partial t} + \frac{\partial^2}{\partial x^2} \left(EI(x) \frac{\partial^2 y}{\partial x^2} \right) = f(x, t) - \sum_{i=1}^n \text{sign}(|x - x_i|) m_i \ddot{y}_i \quad (1)$$

$$m_i \ddot{y}_i + c_i (\dot{y}_i - \dot{y}(x_i)) + k_i (y_i - y(x_i)) = 0 \quad (2)$$

where x is the coordinate along the bridge; y is the vertical displacement of the bridge deck; $\bar{m}$, c and EI are the mass per unit length, damping and the flexural rigidity, respectively; m_i , c_i and k_i are the mass, damping and stiffness of the i th sub-TMD; n is the number of sub-TMDs; x_i and y_i are the location and displacement of the i th sub-TMD; $f(x, t)$ is the vortex-induced force acting on the bridge.

In general, the VIV in bridges is dominated by a single mode, i.e., $y = \phi(x)q(t)$. Substitute $y = \phi(x)q(t)$ into Eqs. (1) and (2), and left multiply ϕ^T at each side of Eq. (1), and integrate from 0 to L , then the motion equations can be expressed as

$$\ddot{q} + 2\xi_s \omega_s \dot{q} + \omega_s^2 q = \frac{1}{M_s} \left(F(t) - \sum_{i=1}^n \phi(x_i) m_i \ddot{y}_i \right) \quad (3)$$

$$\ddot{y}_i + 2\xi_i \omega_i (\dot{y}_i - \phi(x_i) \dot{q}) + \omega_i^2 (y_i - \phi(x_i) q) = 0 \quad (4)$$

where q and ϕ are the generalized coordinate and eigenvector of the bridge deck; ξ_s , ω_s and M_s are the modal damping ratio, frequency and generalized mass of the bridge; $F(t) = \int_0^L \phi(x) f(x, t) dx$ is the generalized force corresponding to the vortex-induced force acting on the bridge deck, in which L is the bridge length; ξ_i , ω_i are the damping ratio and frequency of the i th sub-TMD.

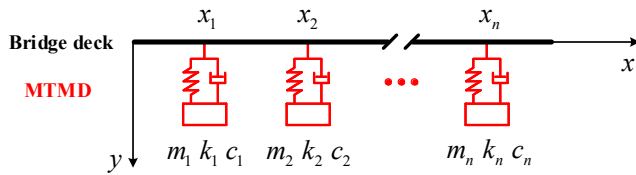


Fig. 1. Bridge-MTMD system.

2.2. Displacement of the bridge and sub-TMD under vortex-induced force

The semi-empirical linear model proposed by Scanlan [44] is used to simulate the vortex-induced force on the bridge, which consists of harmonic force, aerodynamic damping and aerodynamic stiffness, as shown in

$$f(x, t) = \frac{1}{2} \rho U^2 (2B) \left[Y_1 \frac{\dot{y}}{U} + Y_2 \frac{y}{B} + \frac{1}{2} C_L(x) \sin(\omega t + \theta) \right] \quad (5)$$

in which ρ is the air density; U is the mean wind velocity; B is the feature size of the bridge deck; Y_1 , Y_2 and C_L are the aerodynamic parameters and can be obtained by the wind tunnel test; ω and θ are the vortex-shedding frequency and initial phase. It can be found that the coefficients of the aerodynamic damping term and aerodynamic stiffness term are constant at a certain wind speed. Although the aerodynamic damping in the model will cause fluctuations in structure damping ratio, considering that the damping ratio of a long-span bridge during VIV is generally around 0.3%, the effect on the damping effect of MTMD is almost negligible. As for the aerodynamic stiffness, it will change the structure frequency, however, Y_2 is almost equal to zero in the lock-in region [44]. For simplicity, the linear model in Eq. (5) can be reduced as a harmonic force, and a similar treatment can be seen in existing studies [38]

$$f(x, t) = \frac{1}{2} \rho U^2 B C_L(x) \sin(\omega t + \theta) \quad (6)$$

The corresponding generalized modal force can be expressed as

$$F(t) = \int_0^L f(x, t) \phi(x) dx = \frac{1}{2} \rho U^2 B C_{LE} \alpha_1 \sin(\omega t + \theta) \quad (7)$$

where $\alpha_1 = \int_0^L |\phi(x)| dx$; $\alpha_1 \cdot C_{LE} = \int_0^L C_L(x) \phi(x) dx$, in which C_{LE} is the equivalent lift coefficient.

Since the vortex-induced force has been simplified as a harmonic force, the VIV amplitude can be determined by the magnitude of the response function of the generalized coordinate q . Introduce $q(t) = q_0 e^{j\omega t}$ and $y_i = y_{i0} e^{j\omega t}$, in which $j = \sqrt{-1}$, the relationship between the displacement of the bridge and sub-TMD can be derived from Eq. (4),

$$y_{i0} = \frac{2\xi_i \omega_i \omega j + \omega_i^2}{-\omega^2 + \omega_i^2 + 2\xi_i \omega_i \omega j} \phi(x_i) q_0 = (g_i + h_i j) \phi(x_i) q_0 \quad (8)$$

where

$$g_i = \frac{-\eta_i^2 + 4\xi_i^2 \eta_i^2 + 1}{4\xi_i^2 \eta_i^2 + (1 - \eta_i^2)^2} \quad (9)$$

$$h_i = \frac{-2\xi_i \eta_i^3}{4\xi_i^2 \eta_i^2 + (1 - \eta_i^2)^2} \quad (10)$$

where $\eta_i = \omega/\omega_i$ is the ratio of the vortex-shedding frequency to the frequency of the i th sub-TMD.

Substitute Eq. (8) into the Fourier transform of Eq. (3) yields the frequency response of the generalized coordinate q , as shown in

$$h(j\omega) = \frac{1}{M_s - \omega^2(1 + m_e) + 2\xi_s \omega_s \omega j + \omega_s^2} \quad (11)$$

$$m_e = \sum_{i=1}^n \mu_i (g_i + h_i j) \quad (12)$$

where m_e is the complex effective mass of MTMD; $\mu_i = m_i \phi^2(x_i)/M_s$ is the mass ratio of the i th sub-TMD. Compare Eq. (11) and the transfer function of the uncontrolled structure, it can be found that setting MTMD is equivalent to adding a complex effective mass m_e to the original normalized generalized mass, i.e., the proposed complex effective mass can entirely represent the dynamic effect of MTMD on the bridge. From Eq. (8), the displacement of TMD is not only related to its own parameters, but also to the displacement of the bridge at TMD's location. Assume the displacement of the bridge at the i th sub-TMD's location is $A_i = \phi(x_i) q_0$, then the stroke of the sub-TMD can be obtained

$$|y_{si}| = A_i \sqrt{\frac{\eta_i^4}{(1 - \eta_i^2)^2 + 4\xi_i^2 \eta_i^2}} \quad (13)$$

Since there will be a plateau with a broad frequency bandwidth on the frequency response curve of the displacement of the bridge with optimal non-uniformly distributed MTMD, as shown in Section 4, A_i can be taken as a constant within the frequency range of the sub-TMDs. From Eq. (13), the part inside the radical sign plays a decisive role, assume it as f_1 , and the partial derivative of f_1 with respect to η_i is given by

$$\frac{\partial f_1}{\partial \eta_i} = 4 \frac{\eta_i^3 + (-1 + 2\xi_i^2) \eta_i^5}{(4\xi_i^2 \eta_i^2 + (1 - \eta_i^2)^2)^2} \quad (14)$$

When $\eta_i^3 + (-1 + 2\xi_i^2) \eta_i^5 > 0$, $\partial f_1 / \partial \eta_i > 0$. Considering that the damping ratio ξ_i of the i th sub-TMD in MTMD is generally small, it is approximated that f_1 reaches its maximum when $\eta_i = 1$. The stroke of MTMD can be approximately calculated by

$$|y_{si}|_{\max} = A_i \frac{1}{2\xi_i} \quad (15)$$

3. Derivative-based MTMD optimization process in bridge VIV control

From Section 2.2, the VIV amplitude can be determined by the magnitude of the response function of the generalized coordinate q , so the purpose of the optimization is to find the optimal parameters of MTMD (m_i, c_i, k_i) and minimize the maximum of the objective function.

Since $h(j\omega)$ is complex, the calculation of its amplitude requires a radical operation. For simplicity, the square operation is adopted, and the objective function is changed to $|h(j\omega)|^2$, which is equivalent to the original objective function, that is

$$J_1 = \min \max |h(j\omega)|^2 \Leftrightarrow J = \min \max |h(j\omega)| \quad (16)$$

In the optimization process, the mass of each sub-TMD is considered to be fixed and the variables are the damping ratio ξ_i and frequency ω_i ($i = 1, \dots, n$). Since the damping and stiffness of TMD must be positive, the optimization problem is actually a constrained optimization. However, the constraint is the boundary of the feasible region, and when the initial value is chosen reasonably, the feasible points in the search process will not fall on the boundary, i.e., the constraint can be considered inactive [45]. The original constrained optimization problem can be computed as an unconstrained optimization problem.

For a derivative-based optimization algorithm, the derivative of the objective function is necessary. According to the chain rule, the derivative of $|h(j\omega)|^2$ with respect to the optimization variables λ_i (ξ_i or ω_i) at any load frequency ω can be expressed as

$$\frac{\partial |h(j\omega)|^2}{\partial \lambda_i} = \frac{\partial h(j\omega)}{\partial \lambda_i} \cdot h^*(j\omega) + h(j\omega) \cdot \frac{\partial h^*(j\omega)}{\partial \lambda_i} \quad (17)$$

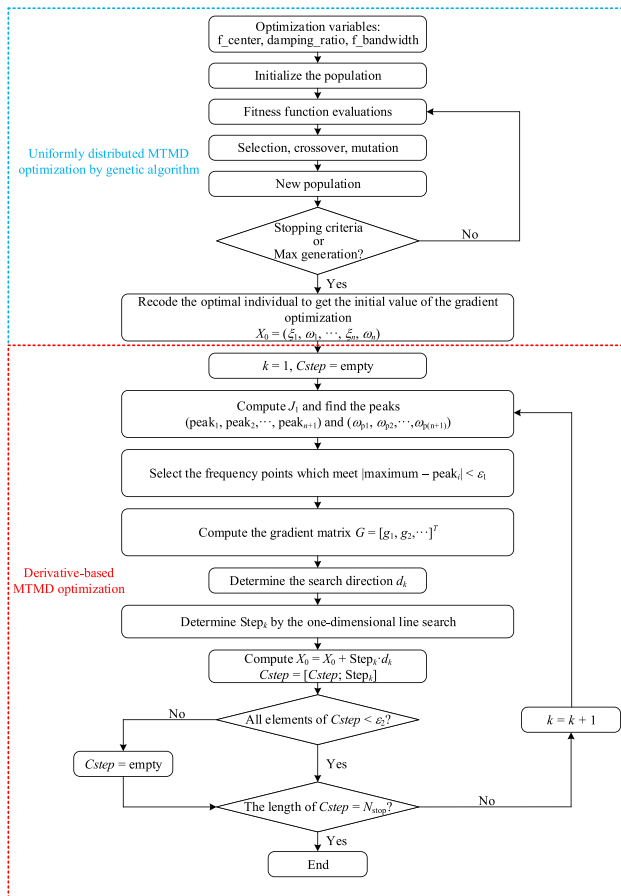


Fig. 2. Flowchart of the proposed optimization algorithm.

Since the derivative operation and the conjugate operation are commutative, it is only necessary to calculate $\partial h(j\omega)/\partial \lambda_i$ and conjugate it to obtain $\partial h^*(j\omega)/\partial \lambda_i$. Also, from Eq. (11), the optimization variables only exist in m_e , and the derivative can be expressed as

$$\frac{\partial h(j\omega)}{\partial \lambda_i} = \frac{1}{M_s} \frac{\omega^2}{[-\omega^2(1 + m_e) + 2\xi_s \omega_s \omega j + \omega_s^2]^2} \frac{\partial m_e}{\partial \lambda_i} \quad (18)$$

where

$$\frac{\partial m_e}{\partial \lambda_i} = \begin{cases} \mu_i \frac{-2\omega_i \omega^3 j}{(-\omega^2 + \omega_i^2 + 2\xi_i \omega_i \omega j)^2} & \lambda_i = \xi_i \\ \mu_i \frac{-\omega^2 (2\xi_i \omega_j + 2\omega_i)}{(-\omega^2 + \omega_i^2 + 2\xi_i \omega_i \omega j)^2} & \lambda_i = \omega_i \end{cases} \quad (19)$$

Up to now, the gradient matrix of $|h(j\omega)|^2$ with respect to the optimization variables $\xi_i, \omega_i (i = 1, \dots, n)$ has been obtained. However, the objective is the maximum of $|h(j\omega)|^2$, therefore, it is necessary to calculate the frequencies ω_p corresponding to the peaks of $|h(j\omega)|^2$. When MTMD consists of n sub-TMDs, there are, in general, at most $n + 1$ peaks, where the frequency of the maximum peak is the target frequency of the optimization, and the search direction of the gradient optimization can be determined by the gradient matrix at this frequency.

It is notable that when there are multiple peaks very close to the maximum, the search direction should consider the gradient of multiple peaks. The method proposed in [29] has been adopted, and the search direction is obtained by averaging the gradient of these peaks according to the angle with the coordinate axes. Moreover, in the process of one-dimensional search, the objective function is not necessarily continuously derivable because the peak corresponding to the objective function may change within a single search step. Although it is numerically continuous, its derivative with respect to the step length does not necessarily exist continuously. Therefore, one-dimensional search methods involving the derivative of the objective function with respect to the step length, such as the Wolfe line search, should be avoided. In this study, the Armijo-Goldstein line search is adopted.

The proposed optimization process is shown in Fig. 2. Firstly, the optimal parameters of MTMD with equal frequency spacing and equal damping ratio are obtained by the genetic algorithm. Then, the optimal uniform MTMD parameters are taken as the initial value of the following derivative-based optimization algorithm, and the target frequency is determined by the find-peaks function. Then the gradient of the target frequency is calculated by Eq. (17)–(19) the search direction can be obtained. Finally, the Armijo-Goldstein line search is used to determine the step length, and the initial value of the next loop can be obtained. Since the convergence speed of the gradient descent method slows down when it is about to approach the local optimal solution, to reduce the computation cost, the optimization procedure is terminated when the search step length is less than ϵ_2 for N_{stop} consecutive times. In this paper, $\epsilon_2 = 1e^{-8}$, $N_{\text{stop}} = 20$. Generally, the optimization results obtained in this way will not be too far from the local optimal solution, and can meet the requirements of parametric analysis and engineering.

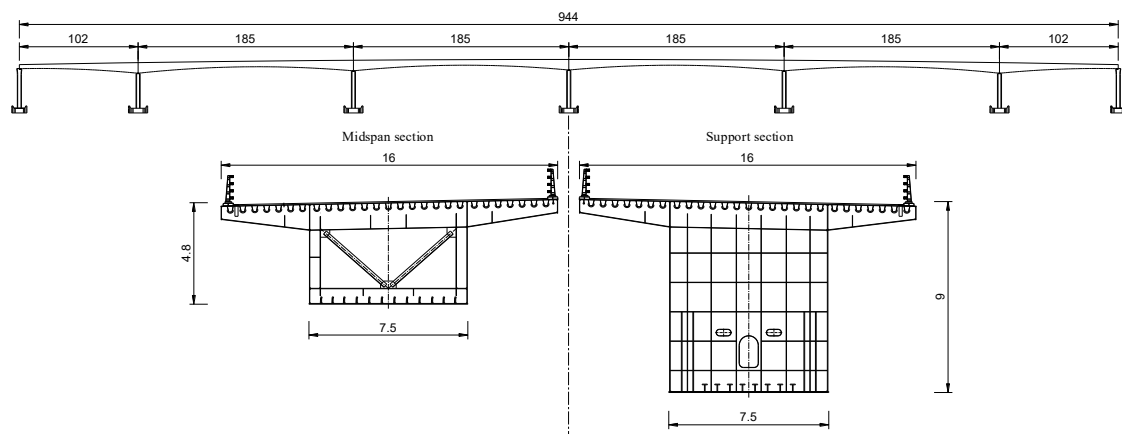
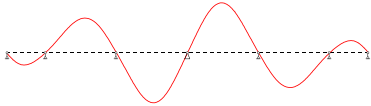
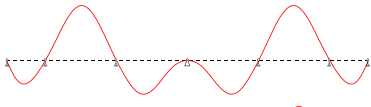
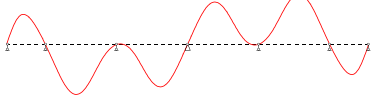
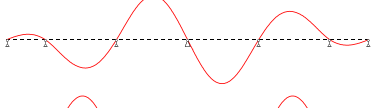
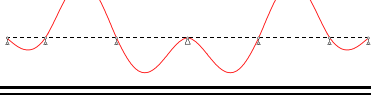


Fig. 3. The continuous steel box-girder bridge layout(unit: m).

Table 1
Modal information of the continuous steel box-girder bridge.

Order	Frequency(Hz)	Generalized Mass(kg)	Mode Shape	Description
1	0.513	3303257.3		1st anti-symmetric vertical bend
2	0.670	3091155.0		1st symmetric vertical bend
3	0.883	3658533.0		2nd anti-symmetric vertical bend
4	0.914	3140186.9		1st anti-symmetric horizontal bend
5	1.129	3104706.5		1st symmetric horizontal bend

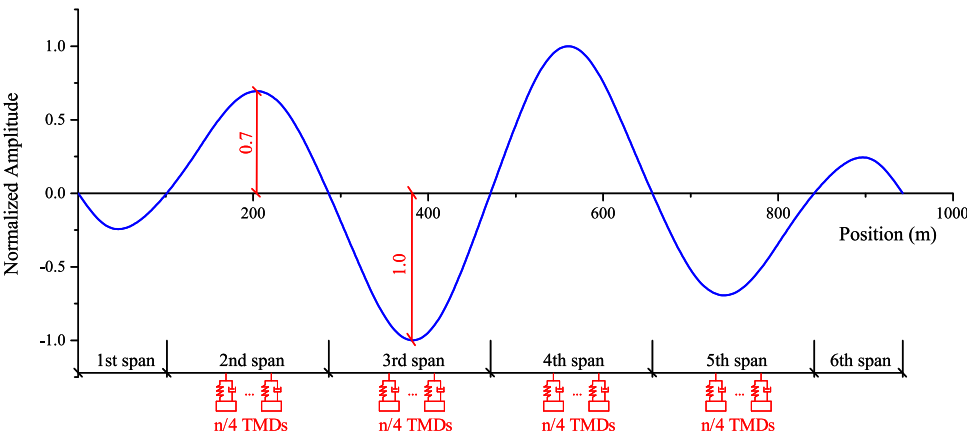


Fig. 4. 1st-order vertical bending mode shape and MTMDs placement positions.

4. Numerical example

4.1. MTMD optimization to mitigate the VIV

A six-span continuous steel box-girder bridge(102+4 × 185+102m) is taken as a numerical example to demonstrate the performance of MTMD for the VIV mitigation in long-span bridges, as shown in Fig. 3. The modal information of the bridge obtained by FEM software ANSYS is shown in Table 1. In this paper, the 1st-order mode is selected as the control mode, which has the possibility of VIV according to the wind tunnel test [46,47] and the mode shape is shown in Fig. 4. The eigenvector component of the 1st-order mode is 1.0 at the midspan of the 3rd and 4th spans, and 0.7 at the midspan of the 2nd and 5th spans. In order to avoid the problems due to the over-high density of dampers, e.g., excessive static deformation, the TMDs are evenly placed at the midspan of the 2nd to 5th spans, and the number of TMDs in each span is equal.

From the Section 2, the bridge with single modal VIV can be simplified as a SDOF system with generalized mass $M_s = 3303257.3\text{kg}$ and circle frequency $\omega_s = 2\pi f = 3.22\text{rad/s}$. Considering that the amplitude of VIV is generally small, the damping ratio of the SDOF

system ξ_s is 0.3%.

As shown in Table 2, three different MTMD layout cases are presented in this section to verify the performance of MTMD for the VIV control in bridges, where case 1: each sub-TMD has equal mass and different frequencies and damping ratios; case 2: the parameters of the sub-TMD at different spans are the same, and the initial value of the derivative-based optimization adopts the uniformly distributed MTMD with equal mass; case 3: the parameters of the sub-TMD at different spans are the same, and the initial value of the derivative-based optimization adopts the uniformly distributed MTMD with equal stiffness proposed by C. Li [48]. For the three cases, the damping ratio of sub-TMD in the initial value is equal, and the mass of each sub-TMD keeps unchanged during the derivative-based optimization process.

It is notable that in the following analysis, TMDs with the same frequency and damping ratio placed in different spans are considered as one TMD, of which the mass is the sum of these TMDs, and the frequency and damping ratio keep unchanged. Therefore, for case 2 and case 3, when the number of sub-TMDs is n , the actual number should be $4n$, but it is assumed that the frequency and damping ratio of the sub-TMD at different spans are the same, so the four sub-TMDs with the same

Table 2

MTMD parameter assumptions of different cases.

Case number	2nd span	3rd span	4th span	5th span	mass of each TMD
1	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	$\frac{m_n}{4} + 1, \frac{\omega_n}{4} + 1, \frac{\xi_n}{4} + 1$ $\vdots$ $\frac{m_n}{2}, \frac{\omega_n}{2}, \frac{\xi_n}{2}$	$\frac{m_n}{2} + 1, \frac{\omega_n}{2} + 1, \frac{\xi_n}{2} + 1$ $\vdots$ $\frac{m_{3n}}{4}, \frac{\omega_{3n}}{4}, \frac{\xi_{3n}}{4}$	$\frac{m_{3n}}{4} + 1, \frac{\omega_{3n}}{4} + 1, \frac{\xi_{3n}}{4} + 1$ $\vdots$ m_n, ω_n, ξ_n	$m_1 = m_2 = \dots = m_n$
2	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	$m_1 = m_2 = \dots = m_n$
3	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	m_1, ω_1, ξ_1 $\vdots$ m_n, ω_n, ξ_n	$m_1 \neq m_2 \neq \dots \neq m_n$

parameters are considered as one. For example, when $n = 1$ (i.e., STMD), there are actually 4 TMDs with the same parameters distributed in four spans. After such processing, the number of sub-TMDs n in the following analysis represents that the set of sub-TMDs has n different frequencies.

With the above assumptions, to ensure the total mass ratio $\mu = \sum \mu_i$ for each case is the same, it is easy to find that the total mass of sub-TMDs in each span is the same, which is $m_t/4$, where m_t is the total mass of MTMD. The total mass ratio of MTMD can be expressed as Eq. (20). At present, a large number of researches have been conducted on the effect of the mass ratio of MTMD on the performance for vibration control, and the results have shown that with the increase of the mass ratio, the performance and robustness are significantly improved [28,29,39,37]. However, considering that it is unrealistic to use an excessively large mass ratio in a bridge, the mass ratio of MTMD is not taken as a variable parameter in this paper, and the total mass ratio $\mu = 1\%$ is selected for all three cases according to some practical engineering [43,1,42].

$$\mu = \frac{\sum m_i \phi^2(x_i)}{M_s} = \frac{m_t \times (0.7^2 \times 2 + 1.0^2 \times 2)}{4M_s} = 0.745 \frac{m_t}{M_s} \quad (20)$$

Table 3Optimal parameters of MTMD for $n = 12$.

No	case 1			case 2			case 3		
	$m_i \phi^2(x_i)$ (kg)	ω_i/ω_s	ξ_i	$m_i \phi^2(x_i)$ (kg)	ω_i/ω_s	ξ_i	$m_i \phi^2(x_i)$ (kg)	ω_i/ω_s	ξ_i
1	1810.51	0.918	0.0088	2752.71	0.921	0.0114	3186.89	0.922	0.0130
2	1810.51	0.930	0.0091	2752.71	0.936	0.0118	3096.79	0.938	0.0128
3	1810.51	0.941	0.0096	2752.71	0.949	0.0117	3010.46	0.952	0.0126
4	3694.92	0.953	0.0135	2752.71	0.961	0.0118	2927.69	0.965	0.0122
5	3694.92	0.968	0.0141	2752.71	0.973	0.0116	2848.28	0.977	0.0118
6	3694.92	0.983	0.0142	2752.71	0.985	0.0115	2772.06	0.990	0.0116
7	3694.92	0.999	0.0145	2752.71	0.998	0.0117	2698.87	1.002	0.0116
8	3694.92	1.015	0.0147	2752.71	1.010	0.0121	2628.53	1.015	0.0117
9	3694.92	1.032	0.0145	2752.71	1.023	0.0123	2560.90	1.028	0.0118
10	1810.51	1.047	0.0107	2752.71	1.038	0.0122	2495.86	1.041	0.0119
11	1810.51	1.060	0.0104	2752.71	1.053	0.0130	2433.26	1.056	0.0122
12	1810.51	1.075	0.0105	2752.71	1.071	0.0137	2372.99	1.073	0.0123

4.2. Performance of VIV mitigation with MTMD

The parameters of MTMD for VIV control of the six-span continuous bridge have been obtained by the optimization process proposed in Section 3. To save space, only the optimal parameters of MTMD for $n = 12$ are listed in Table 3. The normalized frequency response curves of the generalized coordinate of the bridge deck are shown in Fig. 5. It can be seen that tuned mass dampers can significantly reduce the vibration amplitude of the primary structure. Compared with STMD, the maximum of the generalized coordinate of the bridge deck with MTMD is remarkably lower, which indicates that MTMD can provide better control than STMD with the same mass ratio.

For the uniformly distributed MTMD, comparing the performance of the three cases, it can be found that case 3 > case 2 > case 1. The frequency response curve of case 3 is more uniform within a specific frequency bandwidth, mainly because it assumes equal stiffness of MTMD, i.e., $m_i = k/\omega_i^2$, so that the mass distribution of case 3 is actually quadratic along the frequency, and the mass of sub-TMD with lower frequency is larger than that of sub-TMD with higher frequency. Moreover, it can be seen from the curves of case 1 and case 2 that there is an obvious valley in the higher frequency part ($\omega > \omega_s$), indicating that the mass of these two cases is over-distributed in the higher frequency part. Although there is a reduction in the amplitude of the higher frequency part, it does not contribute to the reduction in the amplitude of VIV due to the lock-in phenomenon. The mass of sub-TMD with higher frequency under case 3 is smaller than that of case 1 and case 2, and the mass distribution of case 3 is more reasonable. Therefore, the performance of uniformly distributed MTMD for case 3 is the best.

For the non-uniformly distributed MTMD, it can be seen from Fig. 5 that, when the number of sub-TMDs is small (Fig. 5(a)), the frequency response curve shows multiple peaks with the number of $n + 1$; When the number of sub-TMDs is large (Fig. 5(b)), due to the dense distribution of sub-TMDs around ω_s , the effects of zeros and poles cancel each other out, so that a horizontal line is presented in the frequency response curve instead of a series of peaks. Compared with the uniformly distributed MTMD, the frequency response curve is flatter and the maximum is smaller. For the three cases, there is no difference in the frequency response function maximum for the non-uniformly distributed MTMD.

The maximum of the generalized coordinate of the bridge deck with different numbers of sub-TMDs is shown in Fig. 6. With the same mass ratio of MTMD, the maximum decreases with the number of sub-TMDs. However, the slope of the curve also decreases. It can be seen that when n is greater than 10, the performance of MTMD does not increase significantly by increasing the number of sub-TMDs.

Fig. 7 depicts the frequency distribution and damping ratio distribution of sub-TMDs and the average damping ratio of the optimal MTMD with different numbers of sub-TMDs for the three cases. As the number of sub-TMDs increases, the damping ratio of the optimal MTMD decreases, and the damping ratio of the non-uniformly distributed

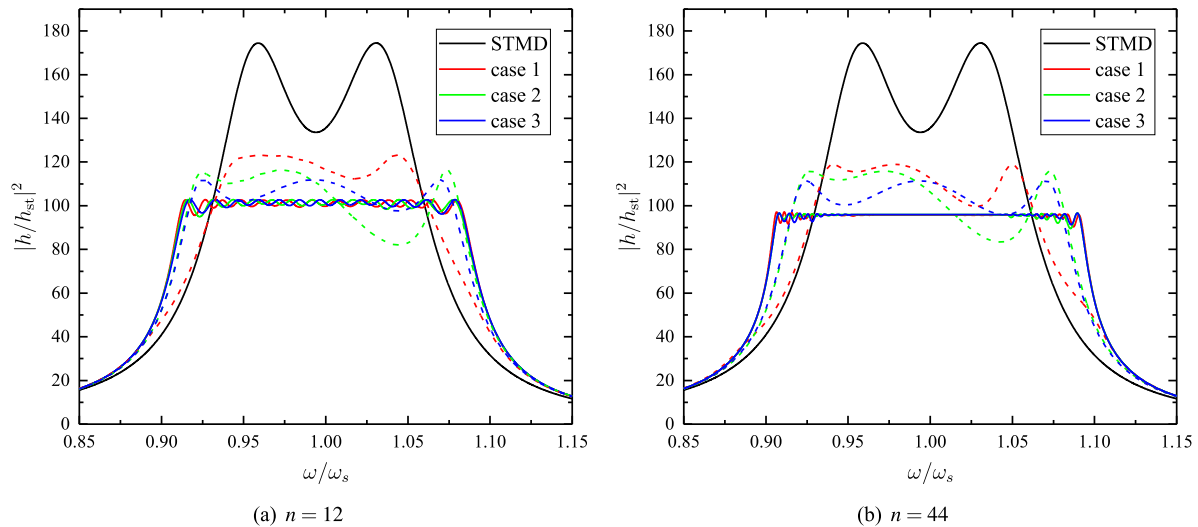


Fig. 5. Frequency response function of the generalized coordinate (Solid line: non-uniformly distributed MTMD. Dashed line: uniformly distributed MTMD).

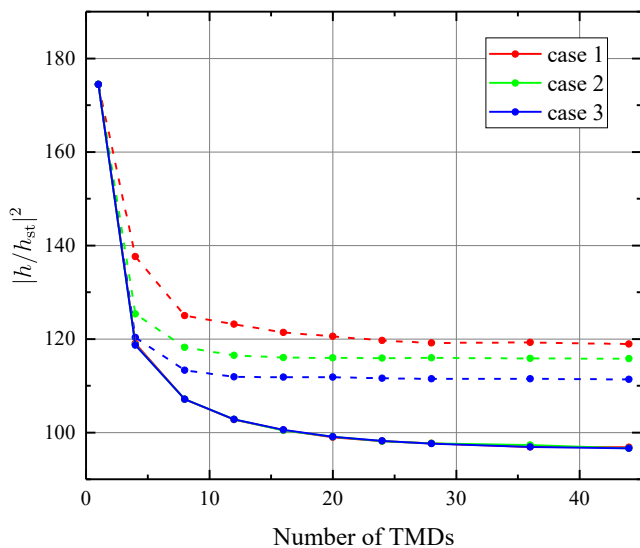


Fig. 6. Maximum of the generalized coordinate with different numbers of sub-TMDs (Solid line: non-uniformly distributed MTMD. Dashed line: uniformly distributed MTMD).

MTMD is lower than that of the uniformly distributed MTMD. It can be seen from Fig. 7(a) that the distribution of sub-TMDs is more intensive near the bridge's eigenfrequency ω_s and the damping ratio distribution of sub-TMDs is approximately symmetric. The damping ratio of the sub-TMD whose frequency is far from ω_s is lower than that of the sub-TMD with a frequency near ω_s . Moreover, as n increases, the frequency bandwidth of MTMD increases, which allows such a series of sub-TMDs with variable frequencies to work on a broader frequency range, so that a plateau is formed on the frequency response curve.

The complex effective mass m_e proposed in Section 2.2 can fully represent the dynamic effect of MTMD on the bridge, as shown in Fig. 8. Since the real part and imaginary part of the effective mass satisfy the Kramers–Kronig relations and can be converted into each other, it is sufficient to focus on one or the other. The imaginary part of the effective mass of MTMD has a larger amplitude than that of STMD. It can be seen from Eq. (11) that the imaginary part of m_e can be merged into the damping term of the host structure to obtain the equivalent damping ratio ξ_e ,

$$\xi_e = \xi_s - \sum_{i=1}^n \frac{1}{2} \mu_i h_i \frac{\omega}{\omega_s} \quad (21)$$

That is to say, the contribution of MTMD is to increase the equivalent damping of the bridge, and the performance of MTMD in VIV control is better due to its larger imaginary part of effective mass. With the increase of n , the amplitude of the imaginary part of m_e becomes larger, and the frequency bandwidth of the frequency response curve becomes broader, as shown in Fig. 5. Furthermore, it can be found that, for the three cases, although the distributions of mass, frequency and damping ratio are different, the curves of m_e almost overlap, so the effects on the bridge are the same. The result also clarifies why the performance of non-uniformly distributed MTMD is the same for the three cases, indicating that the optimal performance of MTMD is independent of the mass distribution of its sub-TMDs.

In order to further analyze the performance and mechanism of MTMD on the VIV control, the time domain analysis of the bridge deck with MTMD has been carried out. The parameters of the bridge are given: $B = 5.5\text{m}$ which is the section height at $L/3$ in the middle span. Since the bridge deck is a bluff body, the Strouhal number is taken as $St = 0.20$ and the equivalent lift coefficient is taken as $C_{LE} = 0.16$ according to the wind tunnel test [47]. The time-history curves of the bridge deck with MTMD are shown in Fig. 9. Since the responses in both frequency domain and time domain are extremely close for non-uniformly distributed MTMD in the three cases, only the result of case 1 is listed, and the curves are vertically offset for clarity in Fig. 9(b). It can be seen from Fig. 9 that the time-dependent displacement curve consists of two stages, the first is from the initial vibration to steady vibration, and the second is the vibration decay stage after unloading. Since the damping ratio of the bridge without control is relatively low, $\xi_s = 0.3\%$, the VIV amplitude is large, and it takes a long time to reach the steady stage. After unloading, the logarithmic decrement $\delta = \ln(A_k/A_{k+1}) = 0.0190$, and the vibration decay is very slow. When tuned mass dampers are adopted, the steady stage amplitude and the time to reach the steady stage are significantly reduced, and the vibration decay speed has improved by an order of magnitude. The results indicate that the introduction of tuned mass dampers has a notable effect on the VIV control of bridges.

Similar to the results of the frequency domain analysis, time domain analysis shows that MTMD has better performance than STMD with the same mass ratio. With the increase of n , the VIV amplitude of the bridge decreases and the decay speed of free vibration becomes faster, but the effect of n on the performance of MTMD will gradually decline. Besides,

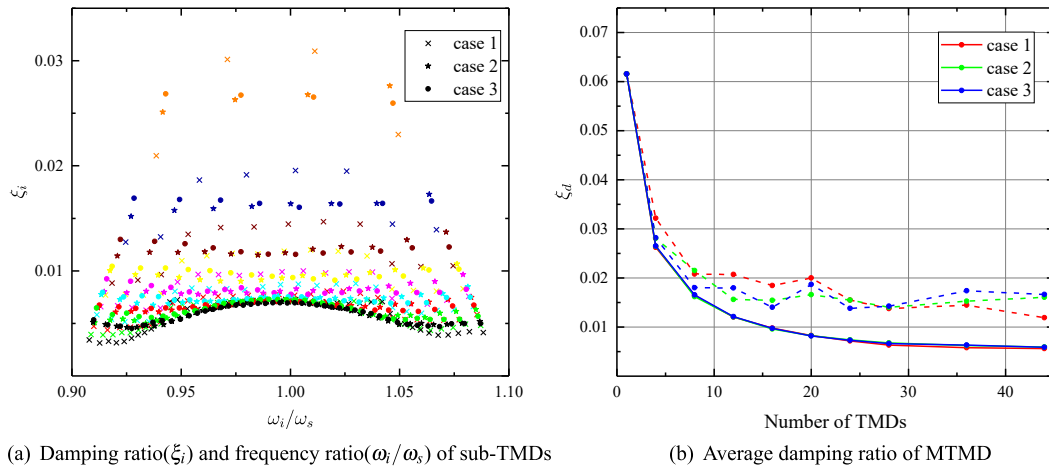


Fig. 7. Damping ratio and frequency ratio of optimal MTMD. (a) Different colors represent different numbers of sub-TMDs(n), n increases from top to bottom; (b) Solid line: non-uniformly distributed MTMD. Dashed line: uniformly distributed MTMD.

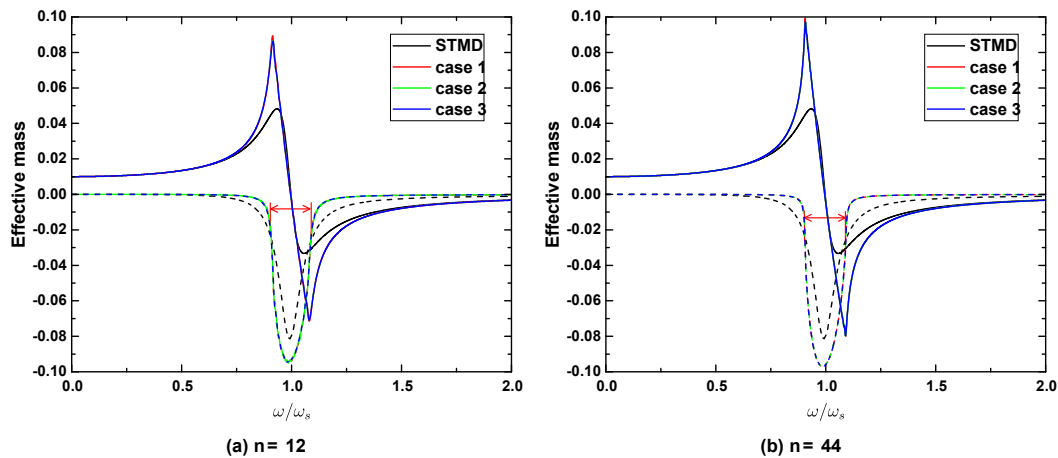


Fig. 8. Complex effective mass of MTMD(Solid line: real part of m_e . Dashed line: imaginary part of m_e).

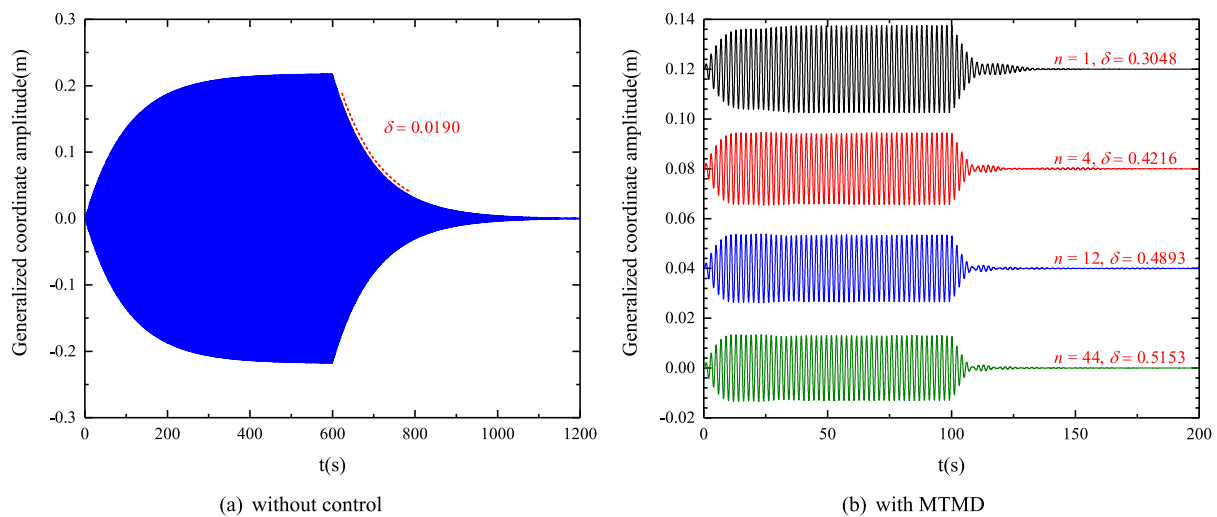


Fig. 9. Displacement history curve of the primary structure.

it is notable that the damping ratio of the optimal MTMD decreases with the number of sub-TMDs, but the logarithmic decrement in the free vibration stage still increases. The strange phenomenon will be analyzed

below.

For the bridge-MTMD oscillation system with $n+1$ degree-of-freedom, the system is expected to have $n+1$ pairs of conjugate eigen-

values, which can be represented by the poles of the transfer function, as shown in Fig. 10. According to the complex mode theory, the displacement of the bridge with MTMD in free vibration can be written in the form

$$x = \sum_{i=1}^{n+1} [\varphi_{1i} y_i(0) e^{\zeta_i t} + \varphi_{1i}^* y_i^*(0) e^{\zeta_i^* t}] \quad (22)$$

where $\zeta_i = \sigma_i + j\omega_{di}$ is the complex eigenvalue of the system; φ_{1i} is the component of the eigenvector in the degree-of-freedom of the bridge generalized coordinate; $y_i(0)$ is the component of the initial displacement in the i th complex eigenmode; the superscript $*$ denotes the complex conjugate operation. Introduce $\varphi_{1i} = \nu_i e^{j\gamma_i}$ and $y_i(0) = T_i e^{j\theta_i}$ into Eq. (22)

$$\text{Damping dissipation energy : } E_{cb} = \int_0^t 2M_s \xi_s \omega_s \dot{q}^2 dt \text{ and } E_{cd} = \sum_{i=1}^n \int_0^t c_i (\dot{y}_i - \phi(x_i) \dot{q})^2 dt \quad (25)$$

$$x = \sum_{i=1}^{n+1} \nu_i T_i e^{\sigma_i t} \cos(\omega_{di} t + \theta_i + \gamma_i) \quad (23)$$

The zeros of the oscillation system are only related to the parameters of each independent sub-TMD, and there is no coupling relation between them. From Fig. 10, the poles and zeros of the system are interlaced, which has been demonstrated for the interlacing property of a lightly damped system in [49]. When the number of sub-TMDs is relatively large, the distributions of poles and zeros in the complex plane are very close. For multiple tuned mass dampers, the damping ratio of sub-TMD is significantly lower than that of STMD, which makes the poles or eigenvalues of the system closer to the imaginary axis in the complex plane, i.e., the real part σ of the eigenvalues is noticeably larger than that of the bridge-STMD system (the poles lie in the left half-plane, and the real part is negative). From Eq. (23), σ_i represents the decay rate of the amplitude of each complex eigenmode, and if the sub-TMDs have the same parameters while the damping ratio is low, it is clear that the fast decay shown in Fig. 9 cannot be achieved. Therefore, the high damping effect exhibited by the bridge-MTMD system does not originate from the exponential terms in Eq. (23), but due to the difference in $\omega_{di}, \theta_i, \gamma_i$, which leads to phase interference between the harmonic terms of each

eigenmode. The interference between the complex eigenmodes of the system is the fundamental reason why the performance of MTMD is better than that of STMD.

For a more in-depth analysis of the mechanism of MTMD and the energy dissipation path of the oscillation system, the energy of each part of the system has been calculated, including the vortex-induced force input energy (E_{input}), the damping dissipation energy of the bridge deck and MTMD (E_{cb}, E_{cd}), and the kinetic energy of the bridge deck and MTMD (E_{kb}, E_{kd}), given by

$$\text{Input energy : } E_{input} = \int_0^t F(t) \dot{q} dt \quad (24)$$

$$\text{Kinetic energy : } E_{kb} = \frac{1}{2} M_s \dot{q}^2 \text{ and } E_{kd} = \sum_{i=1}^n \frac{1}{2} m_i \dot{y}_i^2 \quad (26)$$

Fig. 11 shows the energy history responses of the bridge deck and MTMD with $n = 12$. With the same mass ratio, the bridge deck with MTMD has lower energy input from the vortex-induced force since the vibration displacement of the bridge deck is lower than that of the bridge deck with STMD. It is notable that after unloading, the mass block of STMD also stops moving soon after the bridge deck stops vibrating, whereas, for MTMD, the time scale of its motion is much larger than that of the primary structure. This indicates that although the mass blocks of MTMD remain vibrating for a long period after the primary structure has stopped vibrating, a one-way energy barrier seems to be formed between the bridge deck and MTMD, so that the kinetic energy of MTMD can not be transferred to the primary structure but is dissipated by the damping of MTMD itself.

4.3. Stroke of MTMD

In the practical installation of tuned mass dampers on large-span bridges, the stroke of the mass block is an important design criterion due to the height limitation in the bridge deck. If the stroke is excessive, the mass block may collide with the beam, resulting in the failure of the passive control system or damage to the bridge. Therefore, it is necessary to analyze the stroke of MTMD in the VIV control of bridges.

From the dynamic theory of MTMD for the VIV control proposed in Section 2, the stroke of each sub-TMD can be computed by Eq. (15), and the maximum stroke of sub-TMD is only related to its own damping ratio and the displacement of the bridge at the sub-TMD's location. Each sub-TMD is independent of others and has no coupling. The stroke of MTMD for the three cases is discussed in this part. When VIV occurs in the first vertical bending mode of the bridge, it can be seen from Fig. 4 that, with the same generalized coordinate of the bridge deck, the midspan displacement of the 3rd and 4th spans is larger than that of the 2nd and 5th spans. For case 1, the frequency of the sub-TMD located at the 3rd and 4th spans is closer to the bridge eigenfrequency ω_s than that of the sub-TMD located at other spans. Moreover, from Fig. 7, the damping ratio of the sub-TMD near ω_s is significantly higher than that of the sub-TMD far from ω_s . Therefore, although the displacement of the bridge deck in the 2nd and 5th spans is relatively small, the damping ratio of the sub-TMD placed here is also relatively low, it is necessary to analyze and compare the strokes for all sub-TMDs in this case. It is rather simple

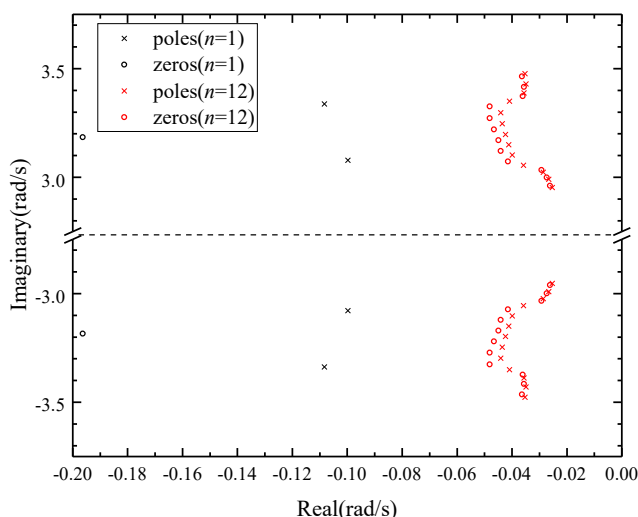


Fig. 10. Poles and zeros of the oscillation system.

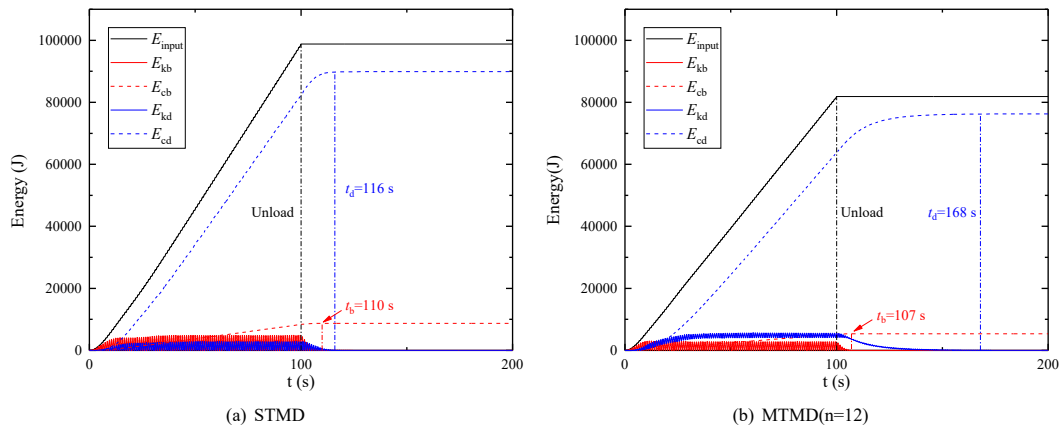


Fig. 11. Energy history responses of the bridge deck and MTMD.

for case 2 and case 3. Because the sub-TMD placed at different spans have the same frequency and damping ratio, the sub-TMD with the highest stroke must be located at the two spans (3rd and 4th) where the displacement of the bridge deck is larger, and only the stroke of the sub-TMDs placed in these two spans is listed in the following results.

The stroke of MTMD for the three cases is shown in Fig. 12. The distribution of the stroke of MTMD with respect to frequency basically corresponds to the distribution of the damping ratio in Fig. 7. The stroke of the sub-TMD near ω_s is smaller than that of the sub-TMD far from ω_s due to their higher damping ratio. Fig. 12(d) shows the variations in the

displacement of the bridge deck and the maximum stroke of MTMD with respect to n . It can be seen that the stroke of MTMD calculated from Eq. (15) is very close to the numerical solution (dashed line), which indicates that it is feasible to estimate the stroke of the optimal MTMD using the proposed simplified formulation. Moreover, unlike the variation of the damping effect with n , the stroke of MTMD increases almost linearly with the increase of n , while the damping effect is inversely proportional to n . Therefore, although a large n can achieve a higher damping effect, the significant increase in the stroke of MTMD makes it unacceptable. For the three cases, it can be found that with the same mass ratio, the

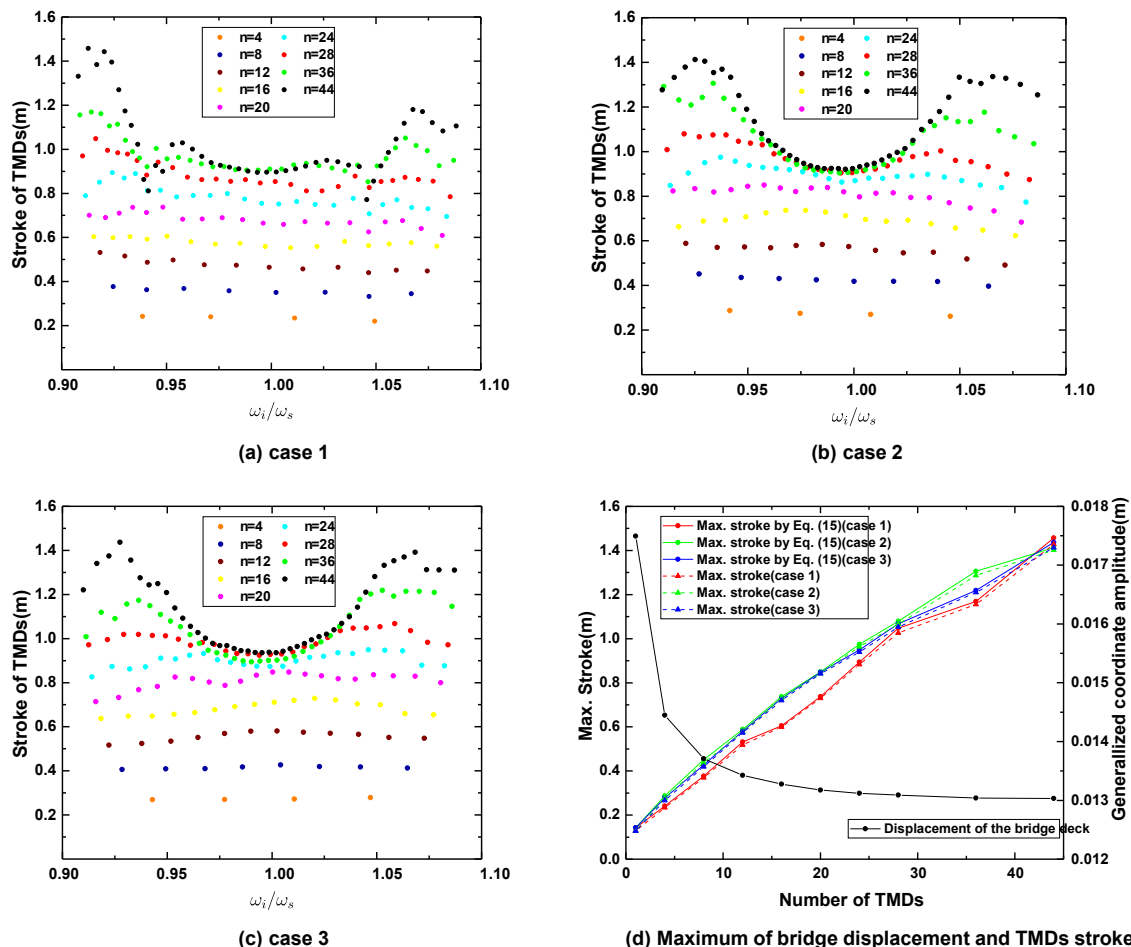


Fig. 12. Strokes of the optimal MTMD.

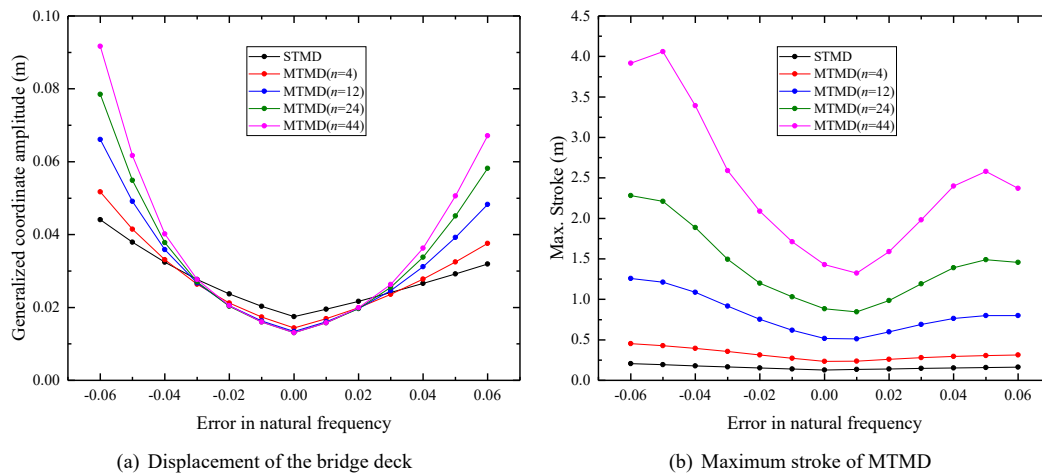


Fig. 13. Energy history responses of the bridge deck and MTMD.

VIV amplitude of the bridge is the same, but the stroke of MTMD for case 1 is the lowest, so case 1 is more recommended in practical applications.

Considering that the damping ratio of MTMD is much lower than that of STMD, the effect of the mistuning phenomenon on the performance of MTMD may be more significant than that of STMD. The sensitivity of the control effect and stroke of MTMD to the error in structural natural frequency is investigated. Due to the presence of mistuning phenomenon, the already obtained MTMD parameters are not optimal for the bridge with error in natural frequency, so Eq. (15) is no longer applicable, and the numerical solution is used for the following analysis. The structural natural frequency error range is taken as -6% to $+6\%$, and the variations in the displacement of the bridge deck and the maximum stroke of MTMD in case 1 to error in structural natural frequency are shown in Fig. 13. It indicates that as n increases, the control effect and stroke of MTMD become more sensitive to error in structural natural frequency. However, as can be seen from Fig. 13(a), although the curve slope of MTMD is larger than that of STMD, the displacement of the bridge deck with MTMD is still lower for an error range of -2.5% to $+2.5\%$. This characteristic of MTMD was also observed by H. Yamaguchi [26] in the analysis of MTMD for suppressing harmonically forced oscillations. For the mass block stroke, as shown in Fig. 13(b), when the error goes from 0 to -6% , the maximum of MTMD increases from 0.2338 m to 0.4533 m ($n = 4$), 0.5182 m to 1.2571 m ($n = 12$) and 1.4282 m to 3.9165 m ($n = 44$), while that of STMD is from 0.1277 m to 0.2065 m. This is reasonable because the damping ratio of MTMD is significantly lower than that of STMD and the sub-TMDs are distributed over a frequency bandwidth, almost symmetrical about ω_s , so that when mistuning occurs, only part of the sub-TMDs are involved and the stroke increases considerably according to Principle of Conservation of Energy. This further illustrates that an excessive n is unrealistic, and n smaller than 12 is more reasonable for practical applications.

5. Conclusion

In this paper, the performance and mechanism of MTMD for the VIV control of long-span bridges was investigated. A complex effective mass was proposed, which can entirely represent the dynamic effect of MTMD on bridges. Based on a semi-empirical linear vortex-induced force model, a derivative-based optimization process has been proposed to determine the optimal parameters of MTMD for the VIV control. A six-span continuous steel box-girder bridge was taken as a numerical example, and a case study for three different MTMD layouts was conducted. According to the results and discussions presented previously, the following conclusions can be obtained:

1. Tuned mass dampers can significantly reduce the displacement amplitude of the bridge deck due to VIV. With the same mass ratio, MTMD has better performance than STMD, and there will be a plateau with a broad frequency bandwidth on the frequency response curve of the generalized coordinate of the bridge with MTMD rather than the peak-valley form which is shown in the STMD scheme.
2. The optimal MTMD with different layouts have the same complex effective mass, and their performance for VIV mitigation is the same, too. It indicates that the optimal performance of MTMD does not rely on the mass distribution of its sub-TMDs.
3. A time-domain analysis was conducted, and it shows that the MTMD scheme with lower damping ratios can still provide a higher damping effect to the host structure. From the perspectives of complex modes and energy dissipation path, the strange phenomenon has been clarified. It reveals the fundamental mechanism of MTMD that the introduction of multiple tuned mass dampers with varied parameters allows the oscillation system to generate multiple complex modes with close frequencies. Through rational tuning, phase interference is generated between the complex modes, forming a one-way energy barrier between the bridge deck and MTMD, which prevents the kinetic energy of MTMD from being transferred to the primary structure to be dissipated only by the damping of MTMD itself.
4. A simplified formula for stroke was proposed, and the stroke of MTMD during VIV was analyzed. It shows that the stroke of MTMD increases almost linearly with the increase of the number of sub-TMDs n while the performance of MTMD for VIV mitigation is inversely proportional to n , indicating that a large n is unrealistic in bridge engineering. Moreover, when there is an error in structural natural frequency, the stroke of MTMD will significantly increase due to its low damping ratio. The robustness of MTMD is a vital issue, including not only the control effect but also the mass block stroke.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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